

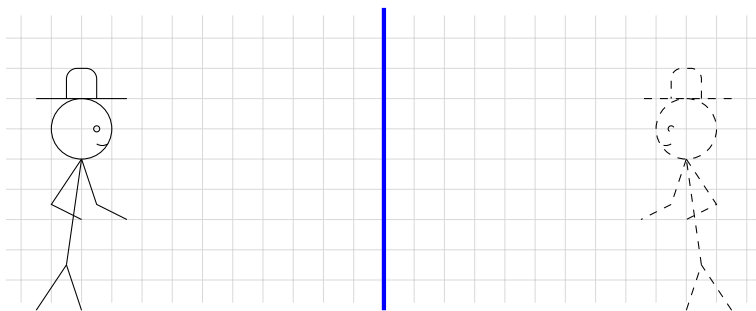
Geometric Optics

The Ray Model of Light

Reflection

The Law of Reflection

The angle of incidence is _____ to the angle of Reflection



Diffuse and Specular Reflection

Curved Mirrors



Name: _____

Date: _____

Period: _____

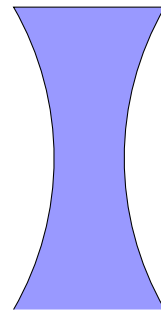
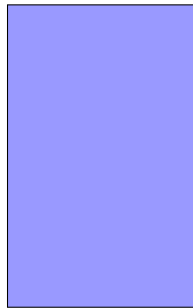
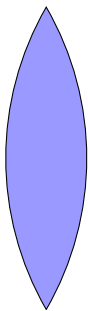
Refraction

- when light slows down, it bends _____ normal.
- when light speeds up, it bends _____ normal.

Total Internal Reflection

- occurs when the angle is _____ for refraction to occur and the light _____ instead.
- occurs when the angle of incidence is larger than the _____.

Lenses



Refraction Practice Problems

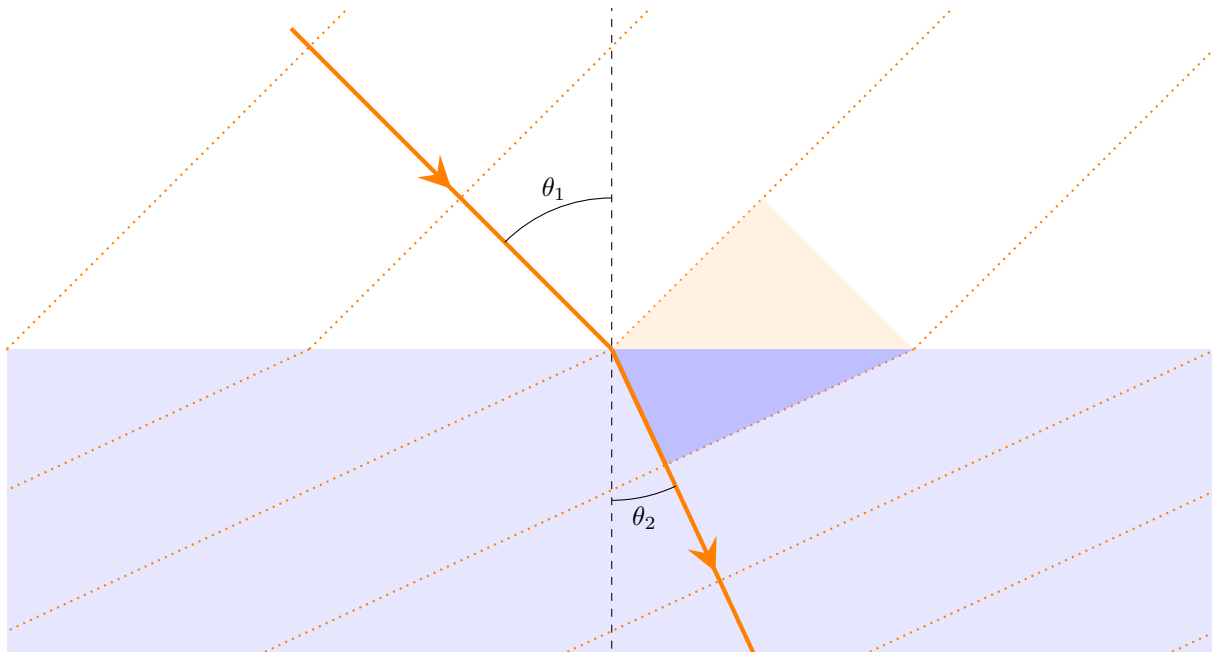
1. The speed of light in ice is 2.29×10^8 m/s. What is the index of refraction of ice?
2. If you shine a laser beam at an angle of 12° to the normal into a block of ice, what is the angle of refraction?
3. A diver shines a flashlight upward from beneath the water ($n = 1.33$) at an angle 35° to the vertical. At what angle does the light leave the water?
4. A flashlight beam strikes the surface of a pane of glass ($n = 1.56$) at an angle of 67° to the normal. What is the angle of refraction?
5. What is the critical angle for the interface between acrylic plastic ($n = 1.49$) and water ($n = 1.33$). To be internally reflected, the light must start out in which medium?

Name:

Date:

Period:

Deriving Snell's Law



Name: _____

Date: _____

Period: _____

Images

An image is the point where light rays _____.

virtual images

rays of light appear to intersect at a location, but in reality there is _____ at that location.

_____ focused on a screen

real images

rays of light really do _____ at a point in space

_____ focused on a screen

Equations for Locating Images

Sign Conventions

d_o distance between mirror/lens and object

d_i distance between mirror/lens and image

f focal length

h_o height of object

h_i height of image

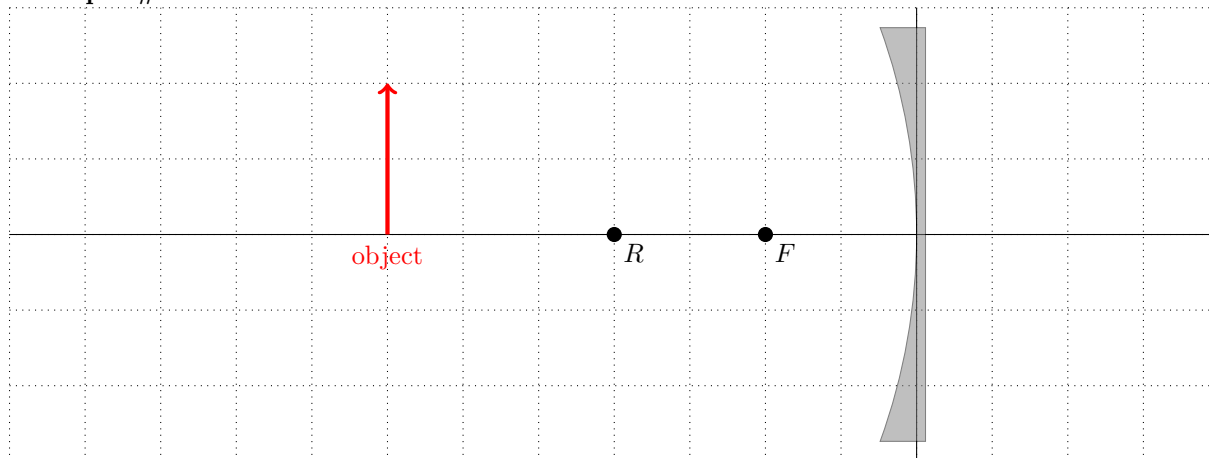
m magnification

Image Formation in Mirrors

Three (Four?) Principle Rays

1. A ray traveling parallel to the principal axis gets reflected to _____
2. A ray traveling through the focal point gets reflected _____
3. A ray that goes through center of curvature gets reflected _____
4. A ray that hits the precise center of the mirror gets reflected _____

Example #1

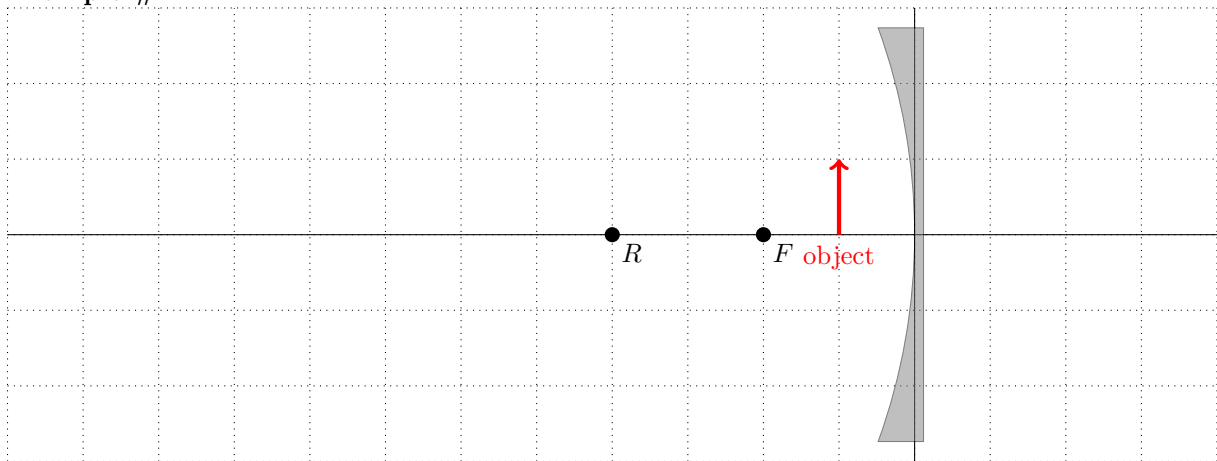


Name: _____

Date: _____

Period: _____

Example #2



Example #3

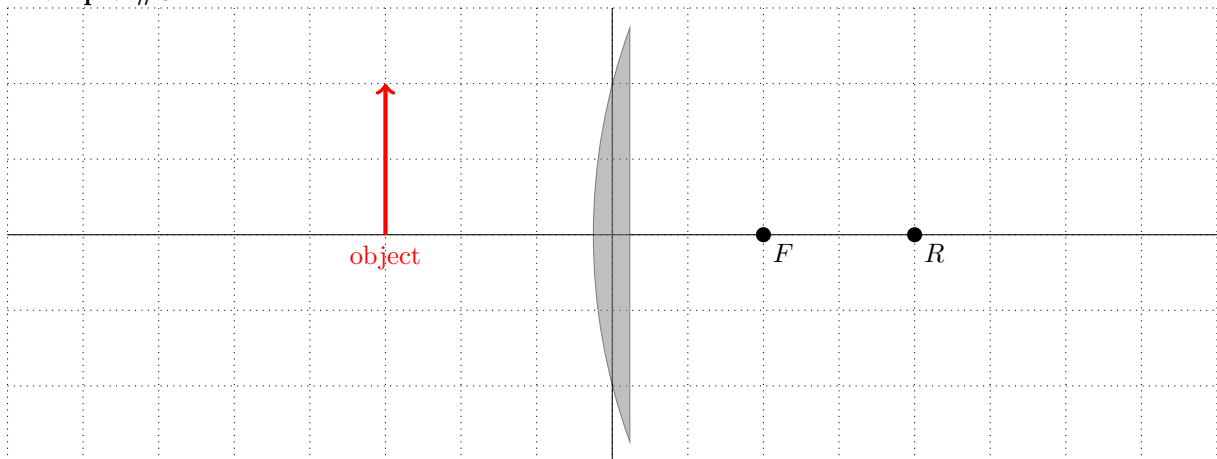
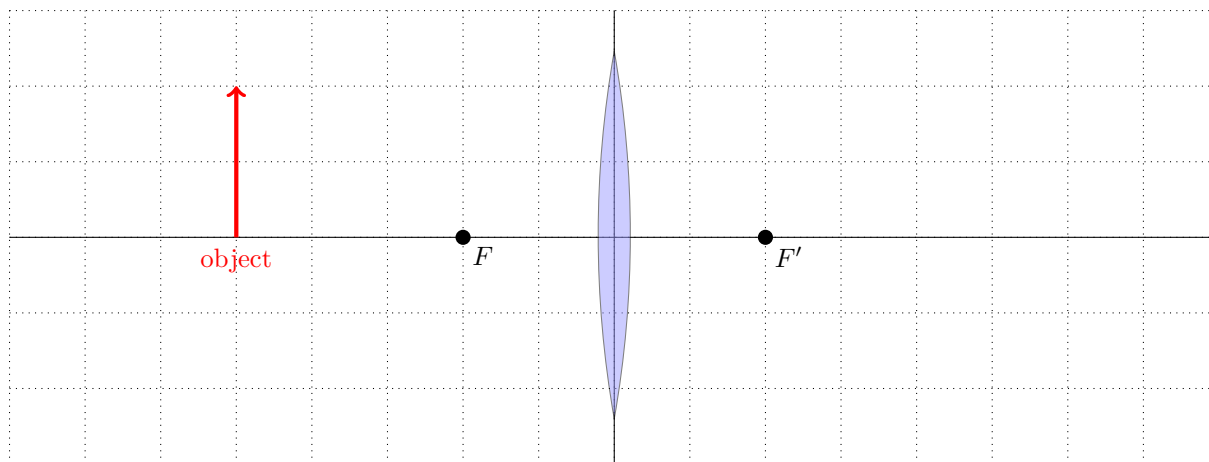


Image Formation in Lenses

Three Principle Rays

1. A ray traveling parallel to the principal axis gets refracted to _____
2. A ray traveling through the focal point gets refracted _____
3. A ray that goes through center of optical center _____

Example #1

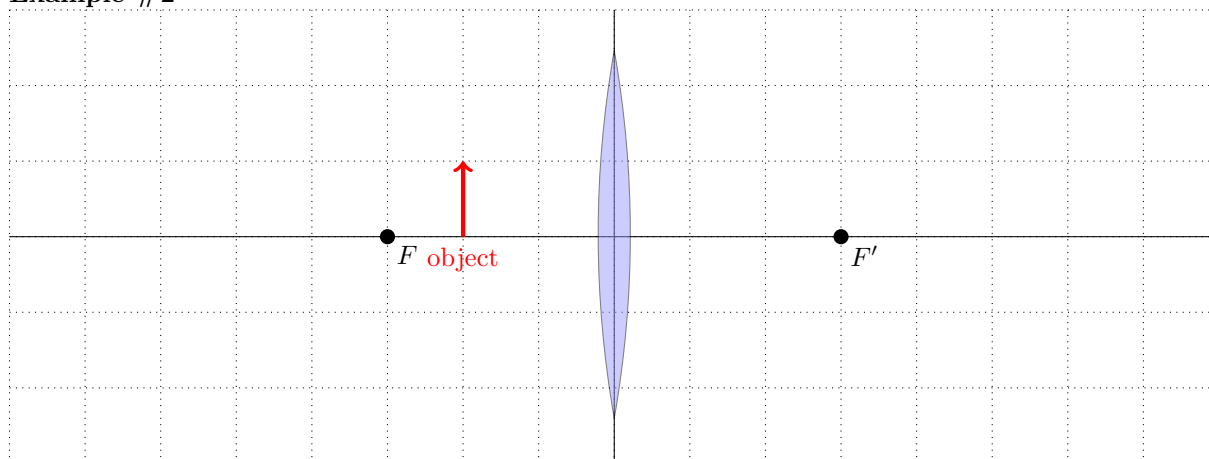


Name: _____

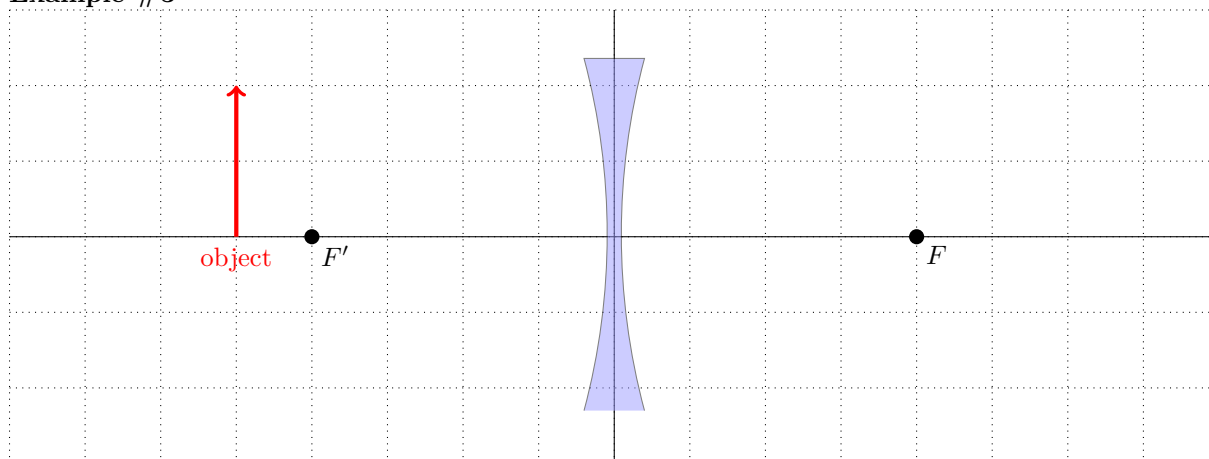
Date: _____

Period: _____

Example #2



Example #3



Lens Equation Practice Problems

1. A rutabaga, which has a height of 44 cm is placed 10 cm in front of a converging lens. The image produced has a height of 66 cm and is *inverted*.
 - (a) What is the image distance?
 - (b) What is the focal length of the lens?
 - (c) What is the power of the lens?

2. A diverging lens has a focal length of 9.0 cm, and an object is placed 3.0 cm from the lens.
 - (a) What would be the distance of the image from the lens?
 - (b) What is the magnification of the image?
 - (c) Will the image be real or virtual, upright or inverted? How do you know?
 - (d) What is the power of the lens?

3. A lens has a power of 0.1 Diopters. Locate the image of an antelope placed upright 30.0 m from the lens. Find the magnification of the image.